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AM-359
Eight-Port Serial I/O Board Installation Instructions

PDI-00359-00, Rev. A01

FIRST EDITION: January 1998

To re-order this document, request part number PDI-00359-00

FCC Notice

This equipment has been tested and found to comply with the limits for a Class A digital device, pursuant to Part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference when the equipment is operated in a commercial environment. This equipment generates, uses and can radiate radio frequency energy and, if not installed and used in accordance with the instruction manual, may cause harmful interference to radio communications. Operation of this equipment in a residential area is likely to cause harmful interference in which case the user will be required to correct the interference at his own expense.

Canadian Department of Communications Compliance Statement

This equipment does not exceed Class A limits per radio noise emissions for digital apparatus set out in the Radio Interference Regulations of the Canadian Department of Communications. Operation in a residential area may cause unacceptable interference to radio and TV reception requiring the owner or operator to take whatever steps are necessary to correct the interference.

Avis de Conformité aux Normes du Ministère des Communications du Canada

Cet équipment ne deapsse pas les limits de Classe A d'émission de bruits radioélectriques pour les appareils numeriques tels que prescrites par le Règlement sur le brouillage radioélectrique établi par le ministère des Communications du Canada. L'exploitation faite en milleu résidential peut entrainer le brouillage des réceptions radio et tele, ce qui obligerait le propriétaire ou l'opératour à pendre les dispositions nécessaires pour en éliminer les causes.

Battery Warning

CAUTION: Danger of explosion if battery is incorrectly replaced. Replace only with the same or equivalent type recommended by the manufacturer. Discard used batteries according to the manufacturer's instructions.
ATTENTION: Il y a danger d'explosion s'il y a replacement incorrect de la batterie. Remplacer uniquement avec une batterie du même type ou d'un type recommandé par le constructeur. Mettre au rébut les batteries usagées conformément aux instructions du fabricant.

For AM-6000 systems, replace batteries with Panasonic or Ray-O-Vac BR1225 only. Use of another battery may present a risk of fire or explosion. Replacement batteries may be ordered from your authorized Alpha Micro reseller.

Electrical Warning

This equipment contains components that can be damaged by static electricity. Follow all electronic discharge precautions when handling the equipment. For example, touch the metal back panel of the CPU or peripheral chassis to dissipate any electrical charge before touching the circuit boards or equipment within the chassis. After turning off power, before you open your computer chassis, unplug the cord from the electrical outlet to guard against electrical shock.

SOFTWARE SECURITY DEVICE IDENTIFICATION NUMBER: _____

The Alpha Micro Software Security Device (SSD) is a customized integrated circuit that personalizes the computer, providing identity verification for it. Certain Alpha Micro and non-Alpha Micro software may require that your computer contain an SSD in order to run software that has been customized to run only on your computer.

Please enter the identification of your SSD above. The SSD identification number should be on your computer ID label under “SSD Serial No.” (Another way of finding the number is to look at the SSD itself. The SSD is located in an integrated circuit location on the CPU board; its identification number is printed on the SSD itself.) Software vendors may ask you for the SSD number if they are customizing software to run only on your computer.

This document may contain references to products covered under the following U.S. Patent Number(s): 4,530,048

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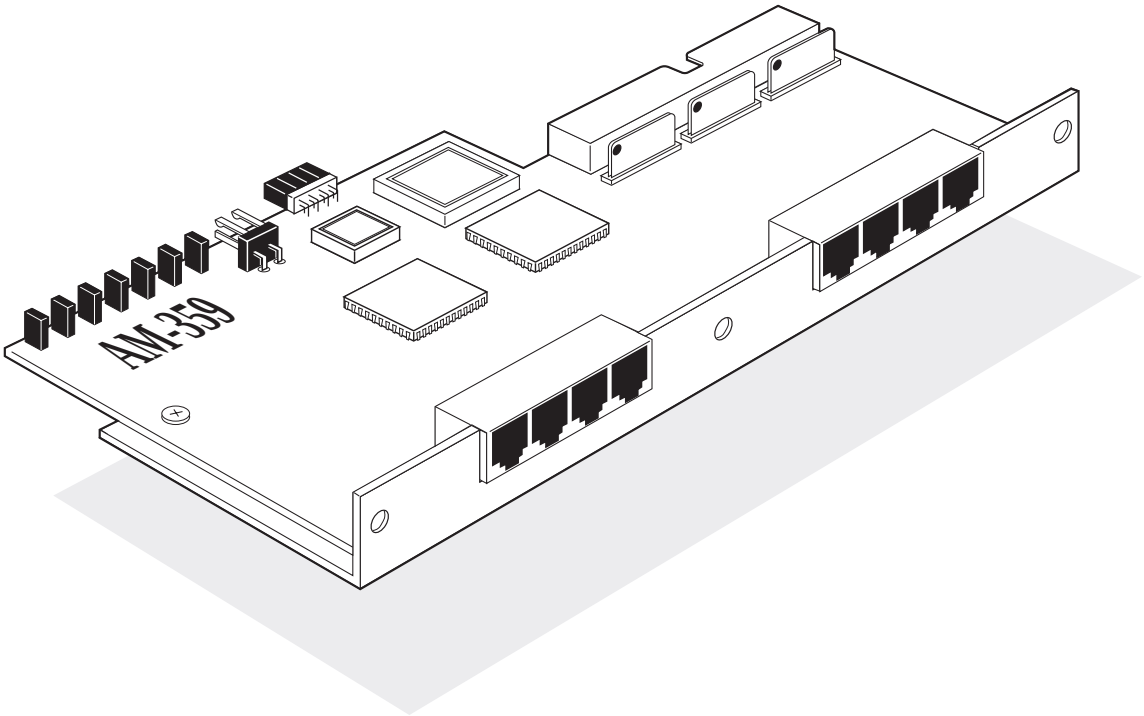
INTRODUCTION

This document gives hardware and software instructions for installing the eight-port AM-359 serial I/O board into Alpha Micro computers that support the 60-pin AM-355 I/O bus.



AM-359(-01) Eight-Port Serial I/O Board

We discuss the following topics:

- General product information.
- Step-by-step hardware installation instructions.
- Installation of multiple I/O boards.
- The procedure for defining one or more AM-359 boards in your system initialization command file.
- Initial testing procedure.

In addition, Appendix A includes information on external cabling required to interface the AM-359 board to external devices; Appendix B contains information on a lightning protection device that can be used in conjunction with the AM-359 board; and Appendix C contains information on the AM-905 I/O expansion chassis.

This document contains information on procedures that should be performed by an experienced service technician. If you are not comfortable performing these procedures yourself, please contact an authorized Alpha Micro service technician for help. Because this product has specific cabling and configuration requirements, the entire document should be reviewed before proceeding with the installation.

GENERAL PRODUCT DESCRIPTION

The AM-359 serial I/O board is a cost-effective way of expanding your computer's serial I/O capabilities. With a single AM-359 board, you can add up to eight serial I/O devices to any AMOS-based computer that supports the 60-pin AM-355 I/O bus. The AM-359 was designed for high speed (up to 57,600 baud), and is compatible with SerialNET under AlphaNET 2.3 (and later).

AM-359 boards support RS232 only; RS422 protocol is not supported.

There are two different versions of the AM-359 board:

- **DWB-00359-01**—the standard AM-359 board with eight RJ-45 eight-wire connectors that support standard RS232 protocol with full hardware handshaking, including modem flow control.
- **DWB-00359-00**—includes all the features included in the -01 board described above, plus the same type of opto-coupling feature as used on Alpha Micro's AM-358 serial I/O board. Opto-coupling uses light-emitting diodes and photo transistors to transfer the signals to and from the remote peripheral devices and the main CPU. Since the communication signals are “opto-coupled,” there is no ground reference to the outside environment, thereby providing ground isolation. In addition to ground loop isolation provided by the opto-coupling, the design provides increased noise immunity from the external environment. The DC/DC isolation on this board avoids potential ground loops created in multi-CPU serial networks.

The AM-359 serial I/O board is designed for the 60-pin AM-355 I/O bus on your computer's main logic board. The AM-350 Intelligent I/O Controller does not support the AM-359 board; however, the use of AM-359 boards will not conflict with any AM-358 or AM-355 boards currently attached to an AM-350 board in your computer.

Maximum Capacities

The AM-359 board uses one-quarter the I/O address space of the eight-port AM-358 serial I/O board. In the same I/O address space used to support a single AM-358 board (eight serial ports), you can install four AM-359 boards (32 serial ports). Because the AM-359 board consumes such a small amount of I/O space (one SIO select address), the number of serial ports you can install is greatly increased.

The list below shows the maximum number of AM-359 boards supported for all system configurations that support the 60-pin AM-355 I/O bus:

The maximum capacities stated below are based on the 60-pin AM-355 I/O bus being populated with only AM-359 boards. Also, your computer may not have mounting locations to support the maximum number of AM-359 boards, in which case an external subsystem designed for I/O expansion is required.

- AM-2000M single board computers support up to 12 AM-359 boards (96 serial ports).
- AM-3000M single board computers, which support dual 60-pin I/O expansion connectors, support up to 28 AM-359 boards (224 serial ports).

- AM-1500, AM-2000, and AM-3000 VME computers support up to 12 AM-359 boards (96 serial ports). VME computers have four less SIO select addresses, 12 compared to 16, than single board computers. This allows I/O space for intelligent controllers like the AM-520 and AM-350 boards.
- AM-4000 computers, which support dual 60-pin I/O expansion connectors, support up to 28 AM-359 boards (224 serial ports).

Mixing I/O Boards

The 60-pin AM-355 I/O bus will support a combination of AM-359, AM-358, and AM-355 serial I/O boards. However, when mixing these boards, you need to address the boards that use the most I/O address space first, followed by the boards that use the least amount of I/O address space. This requires that all eight-port AM-358 boards be addressed first, followed by any AM-355 six-port I/O boards, followed by any AM-359 eight-port I/O boards.

The list below shows all the serial I/O boards supported on the 60-pin AM-355 I/O bus, along with the number of SIO select addresses each board consumes:

1. AM-358 eight-port serial I/O board—four SIO select addresses
2. AM-355 six-port serial I/O board—three SIO select addresses
3. AM-359 eight-port serial I/O board—one SIO select address

Besides serial I/O boards, other types of I/O boards can be connected to the 60-pin AM-355 I/O bus. These boards include the AM-324 parallel printer board, AM-362 Ethernet controller, and AM-333 RJE board. If any of these other types of I/O boards are present in your configuration, you need to insure that you do not create any address conflicts when you install your serial I/O boards. See the section “Using the I/O Address Tables” for more information on mixing various types of boards on the 60-pin AM-355 I/O bus.

Always attach all non-serial I/O boards to the 60-pin connectors located closest to the CPU board, followed by your serial I/O boards.

SPECIFICATIONS AND FEATURES

- Eight RS-232 Serial I/O Ports
- All AM-359 serial ports include full hardware handshaking, including modem flow control signals RTS, CTS, DTR, and DCD.
- Baud Rates—18 rates, software selectable from 50 to 57600 baud.
- All AM-359 serial ports are accessed on the rear panel of your computer via RJ-45 eight-wire connectors.

- The following additional features are available only on the AM-359(-00) board:
    1. DC/DC Power Isolation—Elimination of ground loops.
    2. DC Ground Isolation—External ground and internal ground not connected.
    3. Signal lines opto-coupled to 56K baud for RxD and TxD.
    4. Signal lines opto-coupled to 19.2K baud for RTS, CTS, DTR, and DCD.

Environmental Specifications

Operating Temperature: 60 degrees F to 130 degrees F (16 degrees C to 55 degrees C)

Humidity: 10 to 80% (non-condensing)

Power Specifications

AM-359 Model	+ 5.0 vdc Current	+12.0 vdc Current	-12.0 vdc Current
DWB-00359-01	400 ma	25 ma	10 ma
DWB-00359-00	800 ma	35 ma	0 ma

The AM-905 I/O Expansion Chassis

The AM-905 I/O Expansion Chassis requires a minimum of two I/O boards to maintain a minimum load on the power supply. This can be accomplished by installing two AM-359 boards, two AM-358 boards, two AM-355 boards, or any mixed combination that includes at least two I/O boards. See Appendix C for more information on the AM-905 expansion chassis.

SOFTWARE REQUIREMENTS

The AM-359 board requires AMOS 1.4C or AMOS 2.2C or later. You also need the AM-359 software drivers. AM359.IDV is the driver for the main 60-pin AM-355 I/O bus; AM359B.IDV is the driver used for the “B-Channel” 60-pin I/O connector found on AM-4000 and AM-3000M computers. These drivers are included on all versions of AMOS beginning with the 2/95 Patch Release.

The AM-359 board supports a high-performance feature called Super I/O. Super I/O handles character output much more efficiently than any other previously released AMOS serial port driver, which greatly reduces the load on the CPU and makes more CPU cycles available for other tasks. Under AMOS 2.3 and later, Super I/O is a standard part of AMOS and is enabled by default. See the *System Operator’s Guide to the System Initialization Command File* for information on disabling Super I/O if your situation suggests it.

For instructions on enabling Super I/O under AMOS 2.2C, see *Super I/O Product Support Kit for AMOS 2.2C*, DSS-10539-00. If you have AMOS 2.2C PR 2/95 or later, the Super I/O support files are included; you do not need to order them separately.

INSTALLATION OVERVIEW

This section contains a brief overview of the hardware installation procedures you will follow to install the AM-359 board and cabling into your computer.

Overview:

1. Unpacking the AM-359 board.
2. Configuring the AM-359 board's SIO select jumpers. All other jumper settings are factory preset and should never be changed.
3. Opening the computer.
4. Mounting the AM-359 board.
5. Installation of multiple I/O boards.
6. Installing the internal 60-pin cable and configuring the I/O bus termination resistors.
7. Closing the computer.
8. Defining AM-359 serial ports in your system initialization command file.
9. Installing external cabling options.

Powering Down Your Computer

Before installing the AM-359, perform the following procedures:

1. If your computer contains non-SCSI disk drives, retract the disk heads to protect the disk surface before turning off the computer. To do this, unmount your Winchester drive(s) by typing the following:

   `MOUNT DSKn: /U` RETURN

   Do this for each Winchester logical device, starting with the HIGHEST number (for example DSK3: or DSK2:) and working back to DSK0:.

2. Turn the power switch on your computer to the “off” position and unplug the AC power cord.

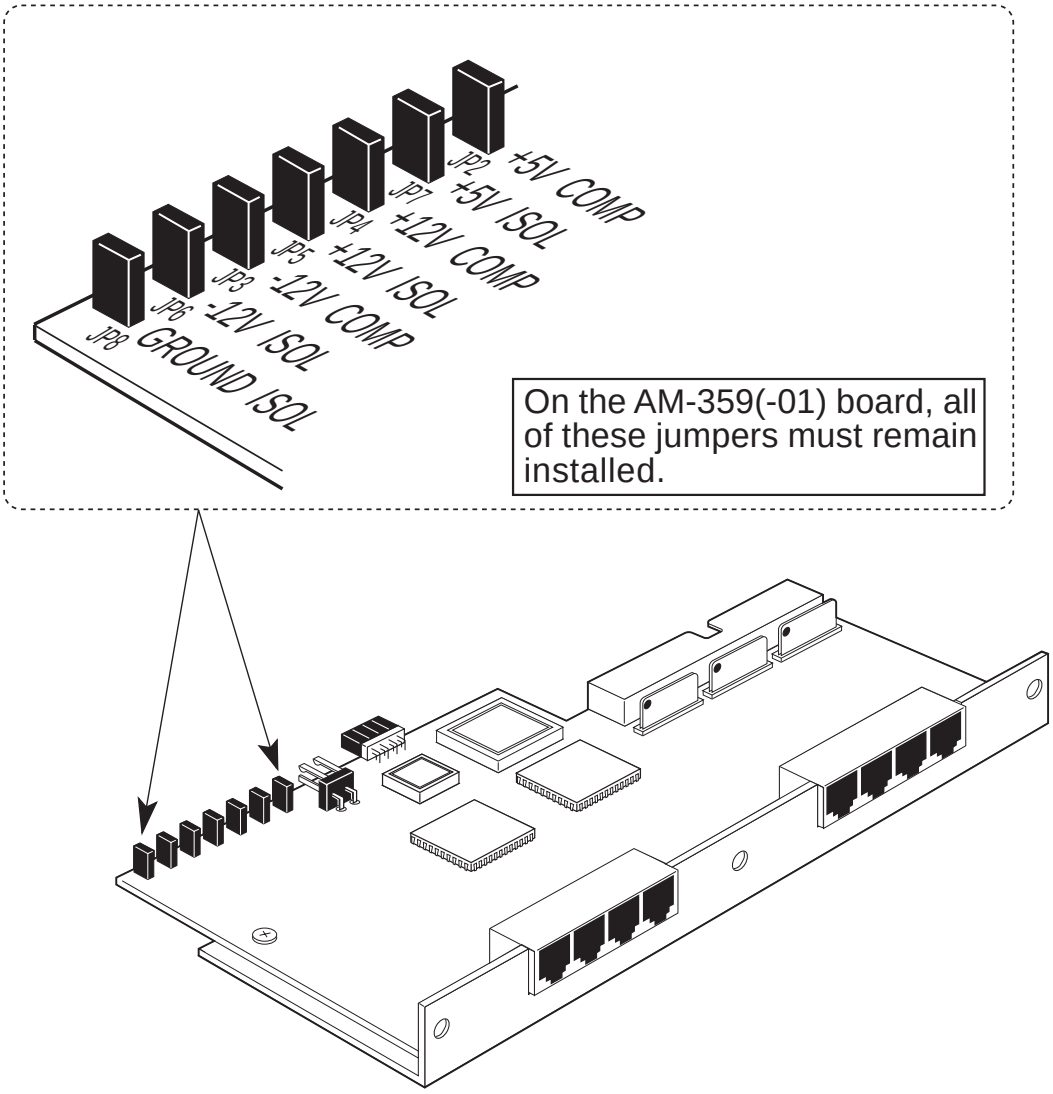
Unpacking

Your body and clothing are capable of storing an electrical charge that can damage or destroy unprotected electronic components. Before handling any computer hardware, make sure your work area is properly protected against static discharge. Do not open the anti-static bag containing your AM-359 board or boards until you are ready to perform the installation.

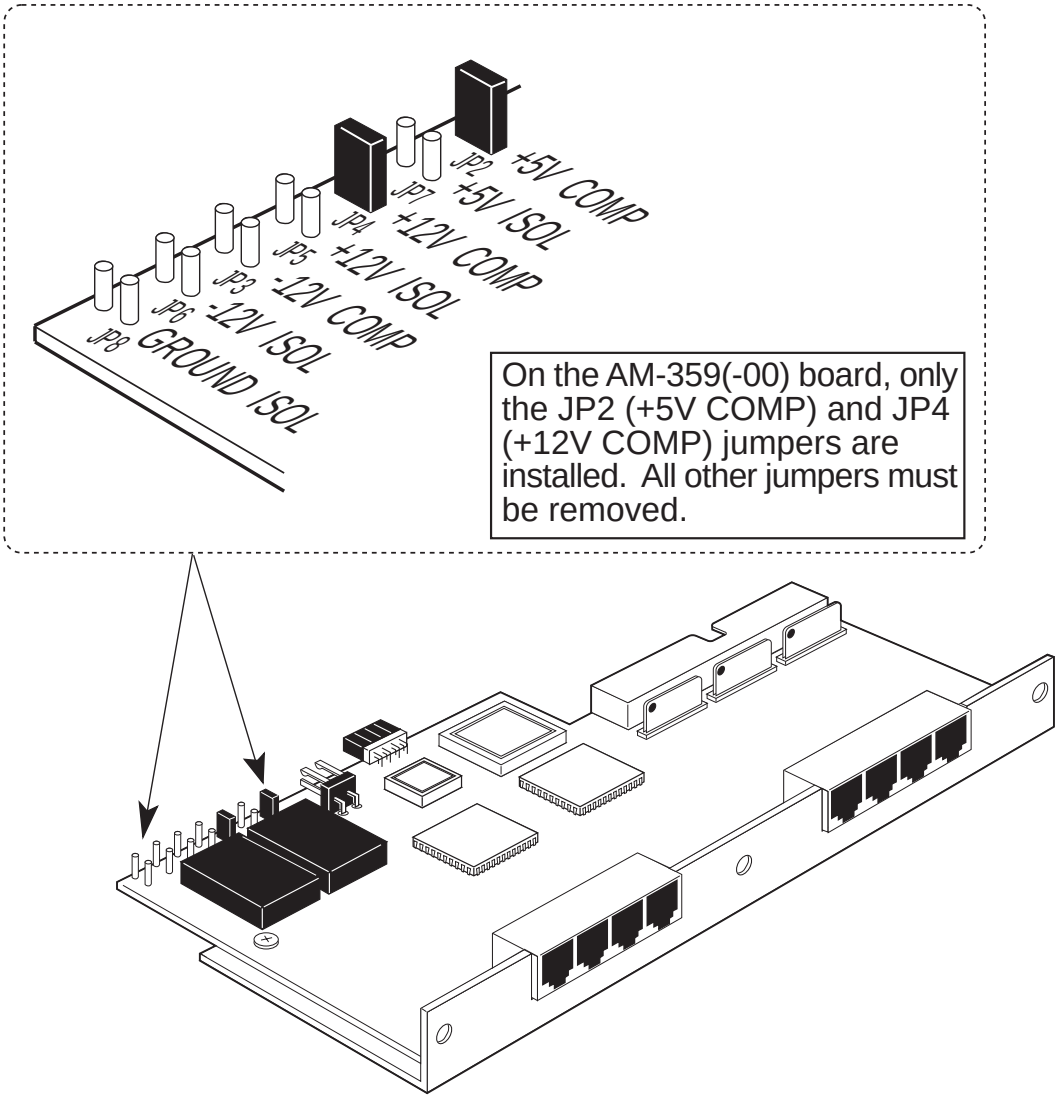
Unpack the AM-359 board or boards. Save the packing material, including the anti-static bag, in case you ever need to transport or return the serial board or boards.

Selecting User Options

The only user configurable jumpers on the AM-359 board are the SIO select jumpers, which are described in the section “Configuring SIO Select Jumpers.” The other jumpers (JP2 through JP8) are factory configured and should never be changed. The following two illustrations show the factory default settings for jumpers JP2 through JP8:



AM-359(-01) Board Default Jumper Settings



AM-359(-00) Board Default Jumper Settings

For information on how to configure the SIO select jumpers, refer to the section “Configuring SIO Select Jumpers.”

Opening the Computer

In order to install the AM-359 board, you need to gain access to the inside of your computer enclosure. Some models have a top cover that must be removed, while others, like the AM-990, have a side panel that must be removed. Complete instructions for opening the computer are located in your computer owner's manual or service manual.

Electronic Equipment Handling Precautions

With the AC power cord unplugged and the top cover removed, the components inside your computer are vulnerable to damage caused by static discharge. Your body and clothing are capable of storing an electrical charge that can damage or destroy unprotected electronic components. Before handling any computer hardware, make sure your work area is properly protected against static discharge. There are a number of commercially available static protection devices, like the wrist strap shown below, designed specifically to protect your equipment from harmful static discharge.

Static Protection Wrist Strap

Mounting the AM-359 Serial I/O Board

The AM-359 mounting procedure is basically the same for all Alpha Micro computers that support the 60-pin AM-355 I/O bus. Each AM-359 board is connected to your computer's rear panel using three 6-32 Phillips-head screws. In AMOS-based AM-990 computers, some serial I/O boards may be attached to the rear panel, while others may be attached to a panel beneath the computer's top cover. However, regardless of where the I/O boards are located, the mounting procedure is the same.

Serial I/O boards should be attached to your rear panel in a logical order. This is accomplished by installing the boards in certain locations based on their SIO select addresses. There are 16 SIO select addresses, ranging from 0 to 15. The serial I/O board with the lowest SIO select address (usually 0) will be installed in the lowest slot on the rear panel, followed by the serial board with the next lowest SIO select address, and so on. By installing serial boards in this manner, the serial ports will be logically arranged on the rear panel.

Always attach all non-serial I/O boards (i.e., AM-362, AM-324, and AM-333) to the 60-pin connectors located closest to the CPU board, followed by your serial I/O boards.

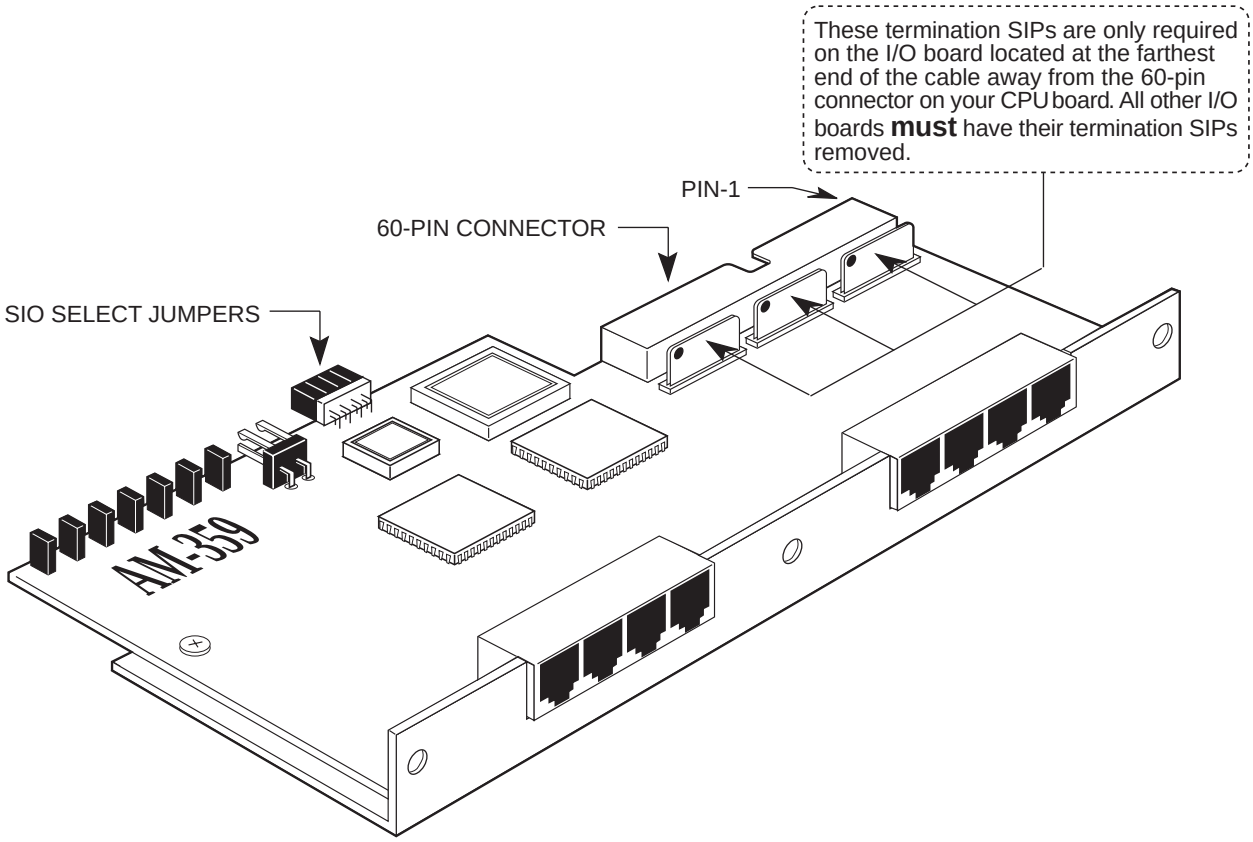
AM-359 Internal Cabling

After mounting all your I/O boards on the rear panel, you must attach the 60-pin flat ribbon cable to each board. This daisy-chain cable links the I/O boards to the 60-pin connector on your CPU board.

On VME computers, it's easier to install the daisy-chain cable when the I/O boards and rear panel assembly are detached from the computer.

Before the release of the AM-359 board, the 60-pin AM-355 I/O bus was limited to no more than five I/O boards. Because the AM-359 I/O addressing scheme is so efficient, a larger number of serial I/O boards can now be installed. To accommodate this, an optional internal cabling kit (PDB-00359-60) is available for configurations where six or more I/O boards will be connected to a single 60-pin cable. For more information about the optional internal cabling kit, see the section, “Special Internal Cabling Options,” later in this document.

Make sure the last I/O board—the one attached to the connector farthest from the CPU board—has the termination SIPs installed. All other I/O boards must have their termination SIPs removed. The termination SIPs are located directly adjacent to the I/O board’s 60-pin connector. The following illustration shows an AM-359 board with the SIPs installed:



Serial I/O Board Termination SIPs

When installing the 60-pin daisy-chain cable, make sure the red stripe on the cable aligns with pin-1 on all the I/O board connectors, as well as pin-1 on the CPU board's 60-pin connector. The illustration above shows the pin-1 location for the AM-359 board's 60-pin connector, which matches the pin-1 location for all of the other boards supported on the 60-pin AM-355 I/O bus. When installed correctly, the 60-pin cable will pass from board to board with no twists.

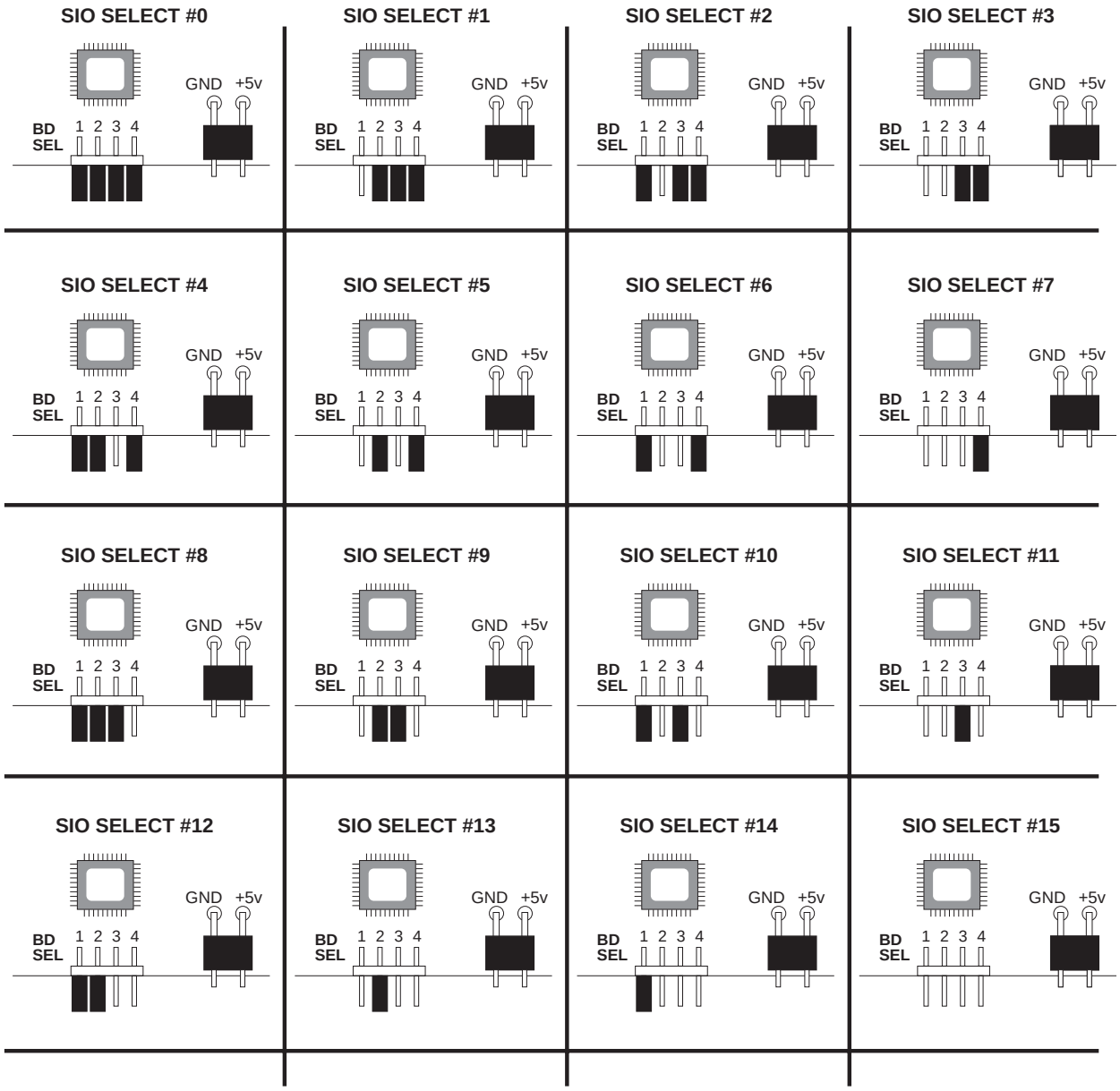
Because the 60-pin cable carries power, it must not be twisted (reversed) as it is sequentially connected to each I/O board. If it is twisted, serious circuit damage will occur.

Multiple I/O Boards

In many cases, you will have more than one type of I/O board connected to your 60-pin AM-355 I/O bus. You may have several types of serial I/O boards, an AM-362 Ethernet card, or even an AM-324 parallel printer board. All these boards can be configured to work together, but you must insure the I/O address space used by each board does not overlap into the I/O space used by another board. When two boards attempt to use the same address space, the resulting condition is called a “conflict.” To determine the best address configuration for your particular I/O application, read the information in the next section, “Configuring SIO Select Jumpers.”

CONFIGURING SIO SELECT JUMPERS

One of the things all I/O boards have in common is their board-select jumpering scheme. The board select jumper area is made up of four jumper positions; 1, 2, 3, and 4. It is a binary select scheme with position 1 being the least significant bit. A jumper installed represents a 0, and a jumper removed represents a 1. There are 16 possible board select numbers: 0 through 15.



I/O Board SIO Select Jumpers

As mentioned earlier in this document, you always want to attach all non-serial I/O boards (i.e., AM-362, AM-324, and AM-333) to the 60-pin connectors located closest to the CPU board. After that, you want to install your serial I/O boards that consume the greatest amount of SIO select numbers (I/O addresses) first, followed by the boards that consume less I/O space. The tables on the next two pages will help you

determine the most efficient method of addressing your serial I/O boards, as well as other I/O boards you may find on the main (also referred to as the A-Channel) 60-pin AM-355 I/O bus. The information immediately following the tables provides additional information about the contents of the tables.

AM-140 and AM-190 CPU boards have an additional 60-pin I/O expansion connector; this extra connector is referred to as the “B-Channel.” More information about the B-Channel can be found later in this section.

Maximum serial port addressing range for AM-355, AM-358, and AM-359 serial I/O boards, based on CPU board type.	I/O ADDRESS	AM355 OCTAL PORT NUMBER	AM358 OCTAL PORT NUMBER	SIO SELECT NUMBERS	ADDRESS CONFLICTS: In addition to serial I/O boards, the AM-355 I/O bus supports other types of I/O boards. This column shows all possible address locations for each type of non-serial I/O board.	AM-359 SIO SELECT NUMBERS AND CORRESPONDING OCTAL PORT NUMBERS
	FF20	0 CPU	N/A	N/A		
	--	1	N/A	N/A		
AM-140 AND AM-145 CPU BOARDS; AM-185 AND AM-190 CPU BOARDS	FC00	2	0	0		SIO Select #0
	--	3	1			Ports 0 - 7
	FC20	4	2	1		SIO Select #1
	--	5	3			Ports 10 - 17
	FC40	6	4	2	AM362 #6	SIO Select #2
	--	7	5			Ports 20 -27
	FC60	10	6	3		SIO Select #3
	--	11	7			Ports 30 - 37
	FC80	12	10	4	AM362 #5	SIO Select #4
	--	13	11			Ports 40 - 47
	FCA0	14	12	5		SIO Select #5
	--	15	13			Ports 50 - 57
	FCC0	16	14	6	AM362 #4	SIO Select #6
	--	17	15			Ports 60 - 67
	FCE0	20	16	7		SIO Select #7
	--	21	17			Ports 70 - 77
	FD00	22	20	8	AM324 #4	SIO Select #8
	--	23	21			Ports 100 - 107
	FD20	24	22	9	AM324 #3	SIO Select #9
	--	25	23			Ports 110 - 117
	FD40	26	24	10	AM324 #2, AM362 #3	SIO Select #10
	--	27	25			Ports 120 - 127
	FD60	30	26	11	AM324 #1	SIO Select #11
	--	31	27			Ports 130 - 137
	FD80	32	30	12	AM362 #2	
	--	33	31			
	FDA0	34	32	13	AM520 (1-4)	
	--	35	33			
	FDC0	36	34	14	AM350 (1-4), AM362 #1, AM333 #1, AM350 (5-8)	
	--	37	35			
	FDE0	--	36	15	AM324 #0, AM333 #0	
	--	--	37			

Table 1. A-Channel I/O Addresses for AM-140/145/185/190 CPU Boards

Each AM-362, AM-324, and AM-333 board shown in column six has a number associated with it. This number has nothing to do with the SIO select number. For example, in column six, located directly adjacent to SIO select 2, you will see “AM362 #6.” The #6 simply identifies it as the sixth possible location for an AM-362 board. There is no connection between the #6 next to AM362 and SIO select 6.

Maximum serial port addressing range for AM-355, AM-358, and AM-359 serial I/O boards, based on CPU board type.	I/O ADDRESS	AM355 OCTAL PORT NUMBER	AM358 OCTAL PORT NUMBER	SIO SELECT NUMBERS	ADDRESS CONFLICTS: In addition to serial I/O boards, the AM-355 I/O bus supports other types of I/O boards. These other I/O boards (i.e., intelligent controller, AM-333, AM-324, and AM-362) use the same I/O address space used by serial I/O boards. Most of these boards have several valid I/O address locations where they can be installed, which helps avoid conflicts. This column shows all possible address locations for each type of non-serial I/O board.	AM-359 SIO SELECT NUMBERS AND CORRESPONDING OCTAL PORT NUMBERS
	FF20	0 CPU	N/A	N/A		
	--	1	N/A	N/A		
	FF40	2 PORTS	N/A	N/A		
	--	3	N/A	N/A		
	FF60	4	N/A	N/A		
	--	5	N/A	N/A		
AM-175 AND AM-180 CPU BOARDS	FC00	6	0	0		SIO Select #0
	--	7	1			Ports 0 - 7
	FC20	10	2	1		SIO Select #1
	--	11	3			Ports 10 - 17
	FC40	12	4	2	AM362 #6	SIO Select #2
	--	13	5			Ports 20 -27
	FC60	14	6	3		SIO Select #3
	--	15	7			Ports 30 - 37
	FC80	16	10	4	AM362 #5	SIO Select #4
	--	17	11			Ports 40 - 47
	FCA0	20	12	5		SIO Select #5
	--	21	13			Ports 50 - 57
	FCC0	22	14	6	AM362 #4	SIO Select #6
	--	23	15			Ports 60 - 67
	FCE0	24	16	7		SIO Select #7
	--	25	17			Ports 70 - 77
	FD00	26	20	8	AM324 #4	SIO Select #8
	--	27	21			Ports 100 - 107
	FD20	30	22	9	AM324 #3	SIO Select #9
	--	31	23			Ports 110 - 117
	FD40	32	24	10	AM324 #2, AM362 #3	SIO Select #10
	--	33	25			Ports 120 - 127
	FD60	34	26	11	AM324 #1	SIO Select #11
	--	35	27			Ports 130 - 137
	FD80	--	--	12	AM362 #2	
	--	--	--			
	FDA0	--	--	13	AM520 (1-4)	
	--	--	--			
	FDC0	--	--	14	AM350 (1-4), AM362 #1, AM333 #1, AM350 (5-8)	
	--	--	--			
	FDE0	--	--	15	AM324 #0, AM333 #0	
	--	--	--			

Table 2. A-Channel I/O Addresses for AM-175 and AM-180 CPU Boards

Each AM-362, AM-324, and AM-333 board shown in column six has a number associated with it. This number has nothing to do with the SIO select number. For example, in column six, located directly adjacent to SIO select 2, you will see “AM362 #6.” The #6 simply identifies it as the sixth possible location for an AM-362 board. There is no connection between the #6 next to AM362 and SIO select 6.

Let's take a look at the information presented in the tables appearing on the two previous pages:

1. The first column shows the maximum address range for serial I/O boards attached to the 60-pin AM-355 I/O bus for each type of CPU board.
2. The second column shows the I/O address associated with each of the 16 possible SIO select numbers. Note, SIO select numbers are not applicable to the on-board CPU serial ports. AM-175 and AM-180 CPU boards have six on-board serial ports. AM-140, AM-145, AM-185, and AM-190 boards have two on-board serial ports. The CPU on-board serial ports have their own reserved address space, which is separate from the address space used by the 60-pin AM-355 I/O bus.
3. The third column shows the AM-355 six-port serial I/O **octal** TRMDEF numbers associated with each SIO select number. One AM-355 board consumes three SIO select numbers. The software driver used for the AM-355 serial ports depends on what CPU board you are using:
   - For AM-175 and AM-180 CPU boards, all AM-355 serial ports and the six on-board CPU serial ports will use the AM355.IDV serial port driver.
   - AM-140, AM-145, AM-185, and AM-190 CPU boards each have their own unique serial port drivers (i.e., AM140.IDV, AM145.IDV, AM185.IDV, etc.). For these CPU boards, all AM-355 serial ports will use the same driver as used by the CPU's two on-board serial ports.
4. The fourth column shows the AM-358 **octal** TRMDEF numbers associated with each of the 16 SIO select numbers. Each AM-358 board consumes four SIO select numbers. In order to be recognized by the computer, only SIO select numbers 0, 4, 8, or 12 can be used when setting the SIO select jumpers on AM-358 boards.
5. The fifth column shows the SIO select numbers. There are 16 valid SIO select numbers (0 through 15). The SIO select numbers enclosed in the rectangle represent the valid AM-358 select numbers. Each SIO select number represents one DUART address.
6. The sixth column includes all non-serial I/O boards (i.e., intelligent controllers, AM-362, AM-324, and AM-333) that may cause address conflicts with your serial I/O boards. Each of the boards in column six is shown positioned across from its valid board select number. To help avoid conflicts, most of the boards have several valid installation locations. Beneath each board is a line and a black dot which shows exactly how much address space each board occupies. By scanning across the tables, you can see what board combinations are in conflict.

   Each AM-362, AM-324, and AM-333 board shown in column six has a number associated with it. This number has nothing to do with the SIO select number. For example, in column six, located directly adjacent to SIO select 2, you will see “AM362 #6.” The #6 simply identifies it as the sixth possible location for an AM-362 board. There is no connection between the #6 next to AM362 and SIO select 6.

7. The last column supplies **octal** port number information for the AM-359 board. One of the first things you will notice is that there are eight octal port numbers associated with each SIO select number. This is possible because all eight serial ports on an AM-359 board only require the address space used by one SIO select number; that's four times less space than an AM-358 serial board requires. As you can see, you can add a large number of AM-359 serial ports into a very small amount of I/O address space.

B-Channel Serial I/O Board Configuration

As mentioned earlier, AM-3000M and AM-4000 computers have a second 60-pin I/O connector called the B-Channel. The B-Channel connector supports only AM-355, AM-358, and AM-359 serial I/O boards. Non-serial I/O boards—i.e., AM-362 and AM-324 boards—are not supported on the B-Channel.

The two 60-pin I/O connectors on AM-4000 and AM-3000M computers are labeled J7 and J8. J7 is the main (A-Channel) connector and J8 is the secondary (B-Channel) connector.

Up to 16 AM-359 boards (128 serial ports) can be installed on the B-Channel; however, the installation of a large number of AM-359 boards may require the installation of an external subsystem designed for I/O expansion.

Serial I/O boards connected to the B-Channel require special serial port drivers:

- All AM-355 serial I/O boards use a driver called AM355B.IDV.
- All AM-358 serial I/O boards use a driver called AM358B.IDV.
- All AM-359 serial I/O boards use a driver called AM359B.IDV.

The B-Channel has its own independent I/O address space, which means there will never be a conflict between other I/O boards in your computer and serial I/O boards connected to the B-Channel.

On the B-Channel, just like the serial I/O boards installed on the main 60-pin I/O connector, you need to address the boards that use the most I/O address space first, followed by the boards that use least amount of I/O address space. This requires that all eight-port AM-358 boards be addressed first, followed by any AM-355 six-port I/O boards, followed by any AM-359 eight-port I/O boards.

The I/O addresses, SIO select numbers, and octal port numbering for the B- Channel are shown in the table below:

I/O ADDRESS	AM355 OCTAL PORT NUMBER	AM358 OCTAL PORT NUMBER	SIO SELECT NUMBERS	AM-359 SIO SELECT NUMBERS AND CORRESPONDING OCTAL PORT NUMBERS
F400	0	0	0	SIO Select #0
--	1	1		Ports 0 - 7
F420	2	2	1	SIO Select #1
--	3	3		Ports 10 - 17
F440	4	4	2	SIO Select #2
--	5	5		Ports 20 -27
F460	6	6	3	SIO Select #3
--	7	7		Ports 30 - 37
F480	10	10	4	SIO Select #4
--	11	11		Ports 40 - 47
F4A0	12	12	5	SIO Select #5
--	13	13		Ports 50 - 57
F4C0	14	14	6	SIO Select #6
--	15	15		Ports 60 - 67
F4E0	16	16	7	SIO Select #7
--	17	17		Ports 70 - 77
F500	20	20	8	SIO Select #8
--	21	21		Ports 100 - 107
F520	22	22	9	SIO Select #9
--	23	23		Ports 110 - 117
F540	24	24	10	SIO Select #10
--	25	25		Ports 120 - 127
F560	26	26	11	SIO Select #11
--	27	27		Ports 130 - 137
F580	30	30	12	SIO Select #12
--	31	31		Ports 140 - 147
F5A0	32	32	13	SIO Select #13
--	33	33		Ports 150 - 157
F5C0	34	34	14	SIO Select #14
--	35	35		Ports 160 - 167
F5E0	--	36	15	SIO Select #15
--	--	37		Ports 170 - 177

Table 3. B-Channel I/O Addresses

Using the I/O Address Tables

At first glance, installing serial I/O boards on the 60-pin AM-355 I/O bus may seem complicated. However, there are really only two important considerations. The first consideration is whether any of the I/O boards are in conflict (attempting to use the same area in the I/O address space). The second consideration, which applies only to serial I/O boards, is determining the correct octal port numbers to use for each serial I/O board.

Octal Port Numbers

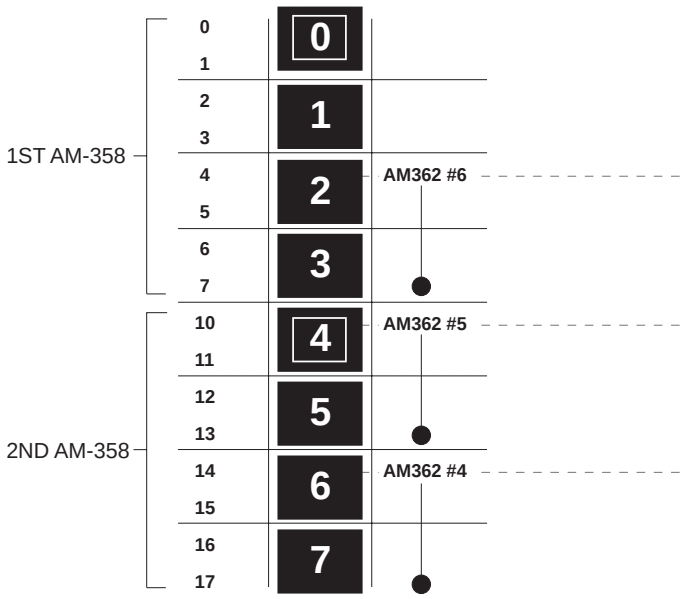
In your AMOS32.INI system initialization command file, each serial port on each serial I/O board is defined using an octal numbering scheme. Each octal number represents one serial port. You might think that mixing serial I/O boards (i.e., AM-355, AM-358, and AM-359) would require some sort of elaborate equation to determine the correct octal port numbering. Actually, just the opposite is true; the octal numbers assigned to each serial I/O board correspond exactly to what SIO select numbers are consumed by each board. This is true regardless of whether you are installing all AM-359 boards or mixing them with other serial I/O boards.

For example:

Let's say you are installing two AM-358 boards followed by one AM-355 board on the main (A-Channel) 60-pin connector on your AM-3000M computer. Since the AM-358 boards consume the most SIO select numbers, they will be installed first. The first AM-358 board will be set to SIO select 0 and consume SIO selects 0, 1, 2, and 3. The second AM-358 board will be set SIO select 4 and consume SIO selects 4, 5, 6, and 7. The AM-355 board will be installed at the next available SIO select number, which is 8. The AM-355 board will consume SIO selects 8, 9, and 10. With the SIO select jumpers properly configured, you must now determine what octal port numbers to use for each of the boards.

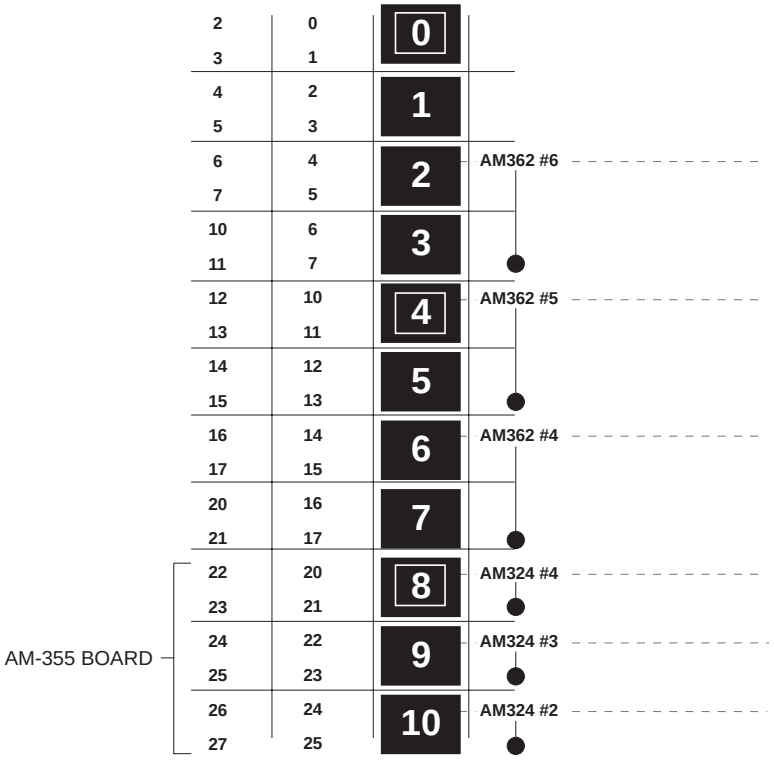
By using the I/O address tables (Table 1 and Table 2 for A-Channel configuration and Table 3 for B-Channel configuration), you can easily determine what octal port numbers to use. Because our three serial I/O boards are attached to the A-Channel on the AM-3000M computer, we'll use Table 1 as our octal port numbering reference source.

If you turn to Table 1, you will see the fifth column contains the 16 SIO select numbers. The AM-358 octal port numbers are located in the fourth column. Because the first AM-358 board consumes SIO selects 0, 1, 2, and 3, the corresponding octal port numbers 0 though 7 will be used. The second AM-358 board consumes SIO selects 4, 5, 6, and 7; if you look at the AM-358 octal port numbers directly adjacent to these SIO select numbers, you can see the second AM-358 board will use octal port numbers 10 though 17. A portion of Table 1 applicable to our AM-358 boards is shown on the next page.



Octal Port Numbers for the Two AM-358 Boards

Just like the AM-358 boards, our AM-355 board uses the octal port numbers corresponding to the SIO select numbers (i.e., 8, 9, and 10) consumed by the AM-355 board. See the illustration below:



Octal Port Numbers for the AM-355 Board

Our example I/O configuration shows the relationship between the SIO select numbers consumed by each serial I/O board and the resulting octal port numbers. This relationship is true when you are mixing different types of serial I/O boards and it is also true when the serial I/O boards are all the same type.

Let's add one AM-359 board to our example. The AM-359 is unique, because all eight serial ports consume only one SIO select number. However, the method for determining which octal port numbers to use for our AM-359 board is exactly the same as used for the other serial I/O boards.

Our AM-355 board consumed SIO selects 8, 9, and 10. This means our AM-359 board will have its SIO select jumpers configured to 11. Using Table 1 as a reference, find the AM-359 octal port numbers in column 7. The table shows that octal port numbers 130 through 137 correspond to SIO select #11.

You do not need to be concerned about the huge gap between the octal port numbers used by the AM-355 board (20 through 27) and the octal port numbers used by AM-359 board (130 through 137). These gaps will always be present when you mix different types of serial I/O boards and they do not cause a problem.

All of the above octal port numbering information can be compressed into one sentence. “The octal port numbers used for each serial I/O board always correspond to the SIO select numbers consumed by each board.”

Using the Tables to Avoid Address Conflicts

Ideally, you want your serial I/O boards to peacefully coexist with other types of I/O boards that may be connected to your 60-pin AM-355 I/O bus. For each serial I/O board you add to your computer, a certain amount of I/O address space is used up and is unavailable for other I/O boards.

For example:

In our sample configuration, which includes two AM-358 boards, one AM-355 board, and one AM-359 board, we used SIO selects 0 through 11. If you look in column six of Table 1, you can see that a number of possible locations for AM-324 and AM-362 boards are contained in the address area used by SIO selects 0 through 11. Because our serial I/O boards have consumed all the I/O address space between SIO selects 0 through 11, no AM-324 or AM-362 boards can be installed in that same area. However, there is still one valid AM-324 address available using SIO select 15 and there are two remaining valid AM-362 locations, SIO select 14 and 12.

Each AM-362, AM-324, and AM-333 board shown in column six has a number associated with it. This number has nothing to do with the SIO select number. For example, in column six, located directly adjacent to SIO select 2, you will see “AM362 #6.” The #6 simply identifies it as the sixth possible location for an AM-362 board. There is no connection between the #6 next to AM362 and SIO select 6.

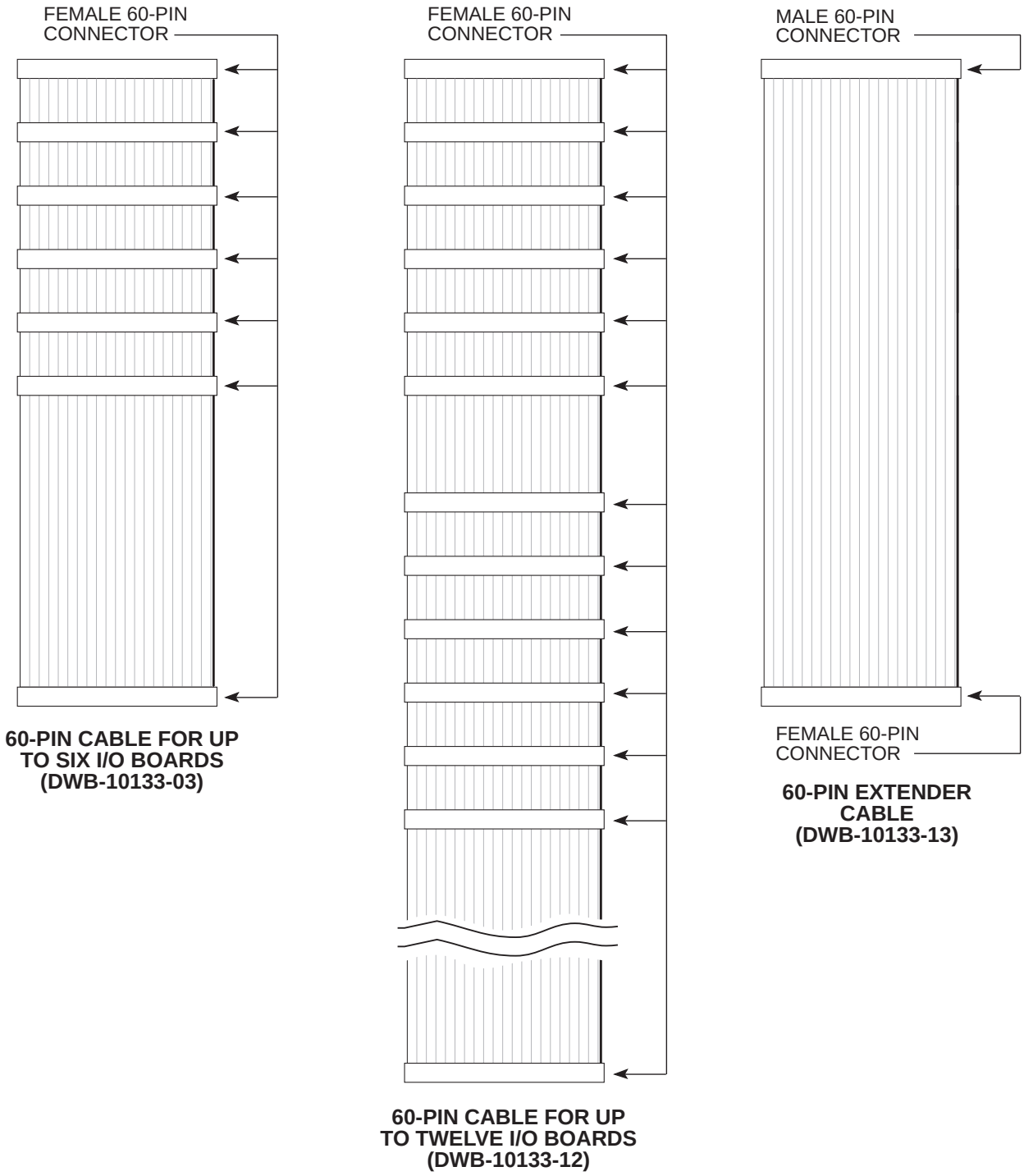
Let's expand our sample configuration and add one AM-324 board and one AM-362 board. We know SIO selects 0 through 11 are being used by our serial I/O boards. This leaves SIO selects 12 through 15 available for our AM-324 and AM-362 boards. By using Table 1, you can see there is only one available location for the AM-324 board, SIO select 15. The area between SIO selects 12 through 15 has two locations for installing AM-362 boards, but only one of these locations will work for our configuration, SIO select 12. The other AM-362 location, SIO select 14, will conflict with our AM-324 board, which

uses SIO select 15. The AM-362 board consumes two SIO select numbers, which means an AM-362 board configured to SIO select 14 would also consume SIO select 15. By choosing SIO select 12 for our AM-362 board, only SIO selects 12 and 13 are consumed, which leaves SIO select 15 available for the AM-324 board.

SPECIAL INTERNAL CABLING OPTIONS

For some configurations, you will simply attach your AM-359 board or boards to your existing 60-pin cable, which has five 60-pin connectors designed for I/O expansion. However, because a large number of AM-359 serial ports can be installed into a very small amount of I/O space, other configurations will require special cables designed to support a large number of AM-359 boards. For these configurations, an optional cabling kit (PDB-00359-60) is available and includes the following:

- One auxiliary +5 volt power cable, which is designed to prevent voltage drops on 60-pin cables with six or more I/O boards.
- One 60-pin cable with six connectors designed for AM-359 and other I/O boards.
- One 60-pin cable with 12 connectors designed for AM-359 and other I/O boards. This cable is designed for the AM-990 double-wide chassis and for older 10 and 21 slot rack mount VME computers.
- One 60-pin extension cable. This cable has one female 60-pin connector and one male 60-pin connector. The female end of the cable plugs into the 60-pin I/O connector on your CPU board. The male end of the extension cable will support either the six or 12 connector I/O cable. By using the extension cable, you can extend the reach of the six or 12 connector I/O cables, which will allow you to install I/O boards in the I/O panel located just underneath the AM-990's top cover.



Optional 60-Pin I/O Cables

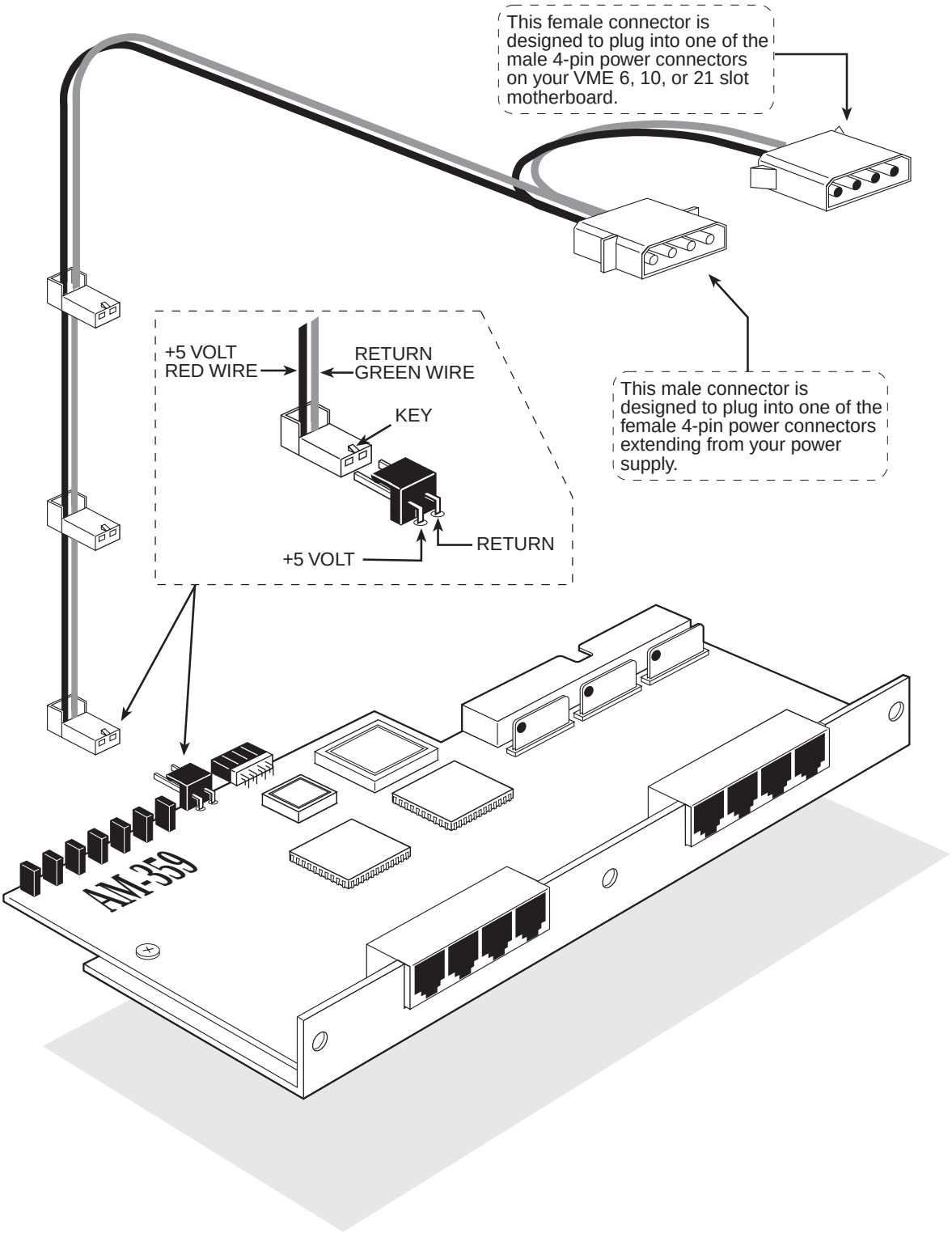
Preventing DC Power Drops on the I/O Bus

The 60-pin cable that links your I/O boards together supplies power to each board. If you have six or more I/O boards connected to your 60-pin I/O cable, you need to plug an auxiliary +5 volt power cable (DWB-10150-02) into the AM-359 board attached to the connector cabled farthest away from the CPU board. The auxiliary power cable supplies an additional +5 volt source and prevents any voltage drop that may occur when six or more I/O boards are connected to the 60-pin cable.

The auxiliary power cable has three +5 volt power connectors designed to plug directly into the AM-359 board. Plug one of the +5 volt connectors into the AM-359 board located farthest away from the CPU board. The remaining +5 volt connectors should be plugged into two other AM-359 boards located closer to the CPU board. The objective is to evenly distribute the +5 volt power between the AM-359 boards on the 60-pin I/0 cable.

In the illustration shown on the next page, you can see three small connectors designed to plug into AM-359 boards. The other two larger connectors (male and female) have 4-pin shells, but only the +5 volt and ground pins are present. The female connector is designed to plug into one of the male 4-pin power connectors on your 6, 10, or 21 slot VME motherboard. The male connector can be plugged into one of the 4-pin power connectors extending from your power supply. ***Only one of these connectors should ever be used*** and when possible, use the male connector. Use the female connector (which is designed to plug directly into your VME motherboard) only if there are no available power connectors extending from the power supply.

The +5 volt connectors that plug into the AM-359 boards are keyed, but if excessive force is used, it is possible to reverse the connection. The +5 volt circuit has high current capabilities and ***if the cable connection is reversed, severe component damage will occur.***



Auxiliary +5 Volt Power Cable Installation

Make sure any unused connectors on the +5 volt auxiliary power cable are properly secured. You want to be absolutely sure that none of these connectors are in a position where they could possibly create an electrical short.

Closing the Computer

When you have finished, install the top cover and rear panels on your computer and fasten them with the original hardware.

DEFINING THE PORTS TO THE OPERATING SYSTEM

Each serial port on each I/O board must be defined in your system initialization command file in order for them to be recognized by your AMOS operating system:

1. A unique name must be assigned to each terminal, printer, or network connected to the individual I/O ports.
2. The correct interface driver must be assigned to each serial port.
3. The octal port number must be defined correctly, as explained in an earlier section.
4. Each port must be assigned a baud rate.
5. An appropriate terminal driver must be assigned, based on the type of terminal you are using.
6. Buffer sizes will vary with different terminal types, but usually 100, 100, 100 is satisfactory.

These definitions are made via TRMDEF statements in the system initialization command file. Below are sample TRMDEF statements for two serial ports:

```
TRMDEF TERM2,AM359=0:19200,AM65,100,100,100   ;(AM-65 Terminal)
TRMDEF TERM3,AM359=1:19200,AM75,100,100,100   ;(AM-75 Terminal)
```

The following sections discuss some of the TRMDEF elements in more detail. Additional information regarding TRMDEF statements and system initialization command files can be found in the *System Operator's Guide to the System Initialization Command File*.

Interface Driver

Two interface drivers support the AM-359 board:

- **AM359.IDV** is the software serial port driver used for all AM-359 boards connected to the main (A-Channel) 60-pin connector on your CPU board.
- **AM359B.IDV** is the software serial port driver used for all AM-359 serial ports connected to the B-Channel on AM-3000M and AM-4000 computers.

Allowable Baud Rates

The AM-359 can be viewed as four independent pairs of serial I/O ports. The baud rates for each pair of serial ports are controlled by a corresponding set of registers in the Quad-UART used on the AM-359. The first set of registers controls the first two ports on the AM-359 board. The next set controls the next two ports on the AM-359 board, and so on. Because of this, the baud rates for the ports in a pair must be compatible with each other. That is, they must come from the same table below:

Table 1	Table 2
50	75
110	110
134.5	134.5
200	150
300	300
600	600
1200	1200
1050	2000
2400	2400
4800	4800
7200	1800
9600	9600
38400	19200

.
57600 baud is supported independent of the baud rate used on any other port.

When assigning baud rates to port pairs, follow these rules:

1. Choose the baud rate you want for the first port of the pair.
2. If this baud rate appears in both tables above—for example, if it's 9600— you can take the baud rate for the second port from either table.
3. If, however, the first baud rate appears only in one table—such as 1050— the rate for the second port must be from the same table.
4. If you assign a baud rate to the second port from the wrong table, the corresponding rate from the correct table is substituted and used instead of the rate you specified.

   For example, you choose 7200 baud for port 0. This selects table #1 for ports 0 and 1. If you then specify a baud rate of 2000 for port 1, which is found only in table #2, the computer substitutes the corresponding rate from table #1, and uses baud rate 1050 for port 1. The baud rates for each pair of ports are independent of any other pair. You may use table #1 for ports 0 and 1, table #2 for ports 2 and 3, and so on.

Modifying the System Initialization Command File

You will need to add a TRMDEF statement to your system initialization command file for each new port. You will also want to modify the JOBS and JOBALC statements to define a job for each new port. Finally, you will add the statements that attach the device for each new port to its job.

The following is a sample initialization command file to verify that the AM-359 I/O board is functional.

This file is created by:

1. FIRST, make a copy of AMOS32.INI by doing the following:

   LOG 1,4 RETURN
   COPY TEST.INI=AMOS32.INI RETURN

   DO NOT MODIFY THE AMOS32.INI file directly. You may be unable to reboot the computer if you make a mistake.

2. View the TEST.INI file:

   VUE TEST.INI RETURN

   Look at the statements in bold in the following system initialization command file for examples of the type of statements you need to add. This system initialization command file is only a sample—the exact contents of the file differ depending on your particular computer.

```
:T
JOBS 25
JOBALC JOB1,JOB2,JOB3,JOB4,JOB5,JOB6,JOB7,JOB8,JOB9,JOB10
JOBALC JOB11,JOB12,JOB13,JOB14,JOB15,JOB16,JOB17,JOB18
TRMDEF TERM1,AM145=0:9600,ALPHA,100,100,100
TRMDEF TERM2,AM145=1:9600,ALPHA,100,100,100
TRMDEF TERM3,AM359=0:9600,ALPHA,100,100,100 ;First AM-359
TRMDEF TERM4,AM359=1:9600,ALPHA,100,100,100
TRMDEF TERM5,AM359=2:9600,ALPHA,100,100,100
TRMDEF TERM6,AM359=3:9600,ALPHA,100,100,100
TRMDEF TERM7,AM359=4:9600,ALPHA,100,100,100
TRMDEF TERM8,AM359=5:9600,ALPHA,100,100,100
TRMDEF TERM9,AM359=6:9600,ALPHA,100,100,100
TRMDEF TERM10,AM359=7:9600,ALPHA,100,100,100
TRMDEF TERM11,AM359=10:9600,ALPHA,100,100,100 ;Second AM-359
TRMDEF TERM12,AM359=11:9600,ALPHA,100,100,100
TRMDEF TERM13,AM359=12:9600,ALPHA,100,100,100
TRMDEF TERM14,AM359=13:9600,ALPHA,100,100,100
TRMDEF TERM15,AM359=14:9600,ALPHA,100,100,100
TRMDEF TERM16,AM359=15:9600,ALPHA,100,100,100
TRMDEF TERM17,AM359=16:9600,ALPHA,100,100,100
TRMDEF TERM18,AM359=17:9600,ALPHA,100,100,100
;
DEVTBL DSK1,DSK2,DSK3,DSK4,DSK5
DEVTBL TRM,RES,MEM
QUEUE 850
BITMAP DSK,,0,1,2,3,4,5
SYSTEM TRM.DVR[1,6]
SYSTEM                          ;No more modules to allocate
SET DSKERR
;
;SETJOB JOB2,TERM2,48K,LOGIN     ;USE SETJOB TO BRING UP PORTS
;SETJOB JOBX,TERMX,48K,LOGIN     ; bring up the rest of the jobs
;
;       Bring up your remaining jobs from here
;
MEMORY 0
```

1. All TRMDEF and SETJOB statements for unused ports should be commented out using the semi-colon (;). See the current *System Operator's Guide to the System Initialization Command File* for more information.
2. Memory allocated for each job depends on job requirements and the amount of memory in the computer.
3. Make sure no one else is using the computer. Then, finish out of the AlphaVUE file by pressing ESC to reach Command Mode. Type **G** and press the RETURN key.

This reboots the computer using the MONTST command. If all lines were input correctly, the computer should reboot. If the computer locks up, press the computer RESET button to reboot the computer under your original system initialization file. Check the TRMDEF statements to be sure you are not trying to access a non-existent channel. Recheck the I/O Configuration Tables and make sure your boards are properly configured.

4. If the system boots successfully, do NOT press the computer RESET button.
5. Make sure all I/O ports are attached as terminal jobs.
6. At the conclusion of boot up, all I/O ports should display the AMOS command prompt.

To make sure the computer recognizes the I/O expansion port, type:

TRMDEF RETURN

to see a list of the terminals defined on the computer.

If you use AMOS 2.2C, see *Super I/O Product Support Kit for AMOS 2.2C*, DSS-10539-00, for information on enabling Super I/O with your AM-359 board. If you use AMOS 2.3 or later, Super I/O is automatically enabled. If, for some reason, you want to disable it on one or more AM-359 boards, see the *System Operator's Guide to the System Initialization Command File*, DSO-00002-00, revision 03 or later.

Finishing the Installation

After you have verified the installation, you can rename your test system initialization command file to its final name. Depending on the original name of your system initialization command file, type:

RENAME/D AMOS32.INI = TEST.INI RETURN

Additional Documentation References

For more information on defining and using serial I/O ports, see the owner's manual that was shipped with your computer. For more general information on adding and changing devices on the Alpha Micro computer, see the *System Operator's Guide* and the *System Operator's Guide to the System Initialization Command File*.

Appendix A - External Cabling Options

There are several different types of plug-and-play external cabling kits available for use with AM-359 serial I/O boards. A part number for each of the cabling kits is shown below along with a brief description of its contents.

The cabling configurations described in this appendix meet the USOC (United Standards Organization Committee) Standards.

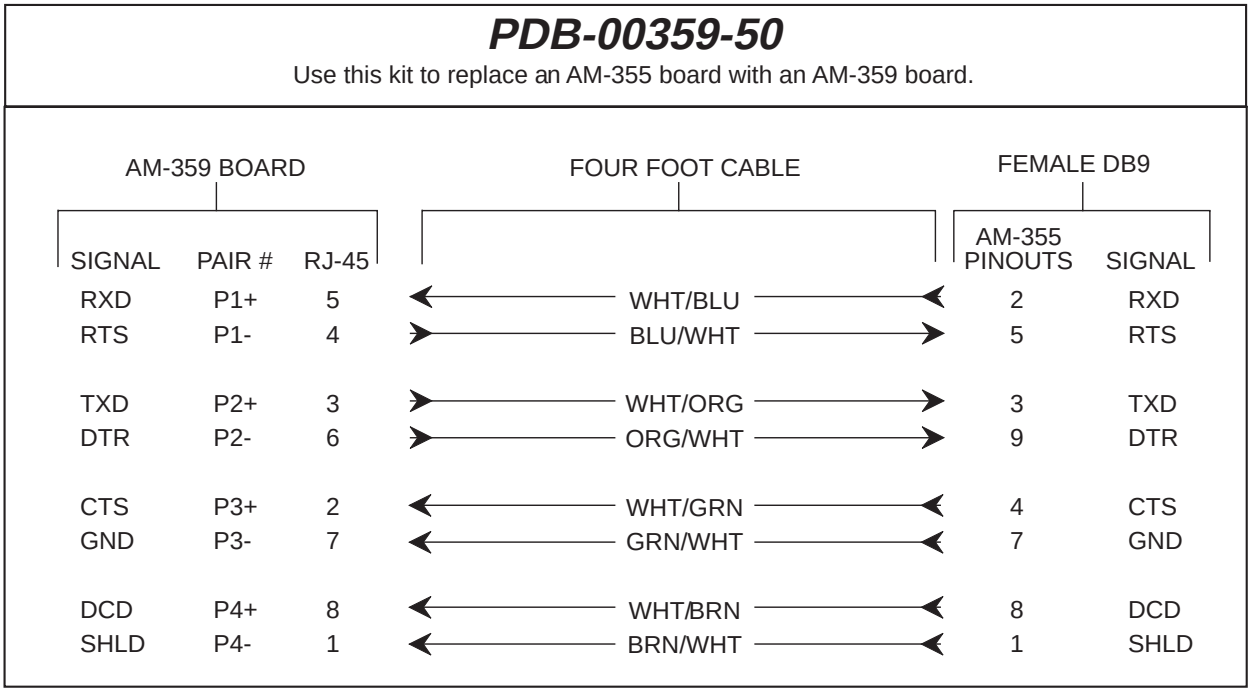
- **PDB-00359-50**—This kit is designed for configurations where an AM-355 board is being replaced with an AM-359 board. The kit includes eight PRA-00243-04 cable assemblies. Each assembly has a four-foot parallel patch cord with an RJ-45 connector at one end and a modular female DB9 connector at the opposite end. To use this kit, you simply plug one RJ-45 connector into each of the AM-359 serial ports and the opposite end of the cables with the female DB9 connectors will accept the male DB9 connectors previously plugged into the AM-355 board.
- **PDB-00359-51**—This kit is designed for configurations where an AM-358 board is being replaced with an AM-359 board. It is compatible with an existing six-wire patch panel using a three-pair wiring system. The cable assembly has eight RJ-45 connectors with six wires each, which plug into the AM-359 board. At the opposite end of the cable assembly, the eight parallel patch cords are tied together using a female RJ-21 connector with bail locks. The female RJ-21 connector will plug into your six-wire patch panel.

This cabling kit does not support DCD for modems, but it does support DTR.

- **PDB-00359-52**—This kit is also designed for configurations where an AM-358 board is being replaced with an AM-359 board. It is compatible with an existing eight-wire patch panel using a four pair wiring system. The cable assembly has six RJ-45 connectors with eight wires each, which plug into six of the AM-359 board's RJ-45 connectors. At the opposite end of the cable assembly, all six parallel patch cords are tied together using a female RJ-21 connector with bail locks. The female RJ-21 connector will plug into your eight-wire patch panel. This kit also includes two of the four-foot long RJ-45 to female DB9 connectors used in the PDB-00359-50 cabling kit.
- **PDB-00359-53**—This kit includes four RJ-45 to male DB25 adapters, which are designed for interconnecting a modem to the AM-359 board. This kit requires the use of a shielded parallel patch cord with RJ-45 connectors at each end, like the 10 foot PRA-00189-10 cable supplied in the PDB-00359-56 cabling kit.
- **PDB-00359-54**—This kit includes four modular RJ-45 to male DB25 adapters, which are designed for interconnecting printers and terminals to the AM-359 board. This kit requires the use of a shielded parallel patch cord with RJ-45 connectors at each end, like the 10 foot PRA-00189-10 cable supplied in the PDB-00359-56 cabling kit.

- **PDB-00359-55**—This kit includes four modular RJ-45 to female DB9 adapters, which are designed for interconnecting a PC-AT style serial port to an AM-359 board. This kit requires the use of a shielded parallel patch cord with RJ-45 connectors at each end, like the 10 foot PRA-00189-10 cable supplied in the PDB-00359-56 cabling kit.
- **PDB-00359-56**—This kit includes four (PRA-00189-10) 10 foot shielded parallel patch cords with RJ-45 connectors at each end.

The following illustrations contain signal and pinout information for each of the PDB-00359-XX cabling kits:



PDB-00359-50 External Cabling Kit

PDB-00359-51

Use this kit to replace an AM-358 I/O board with an AM-359 board when connecting to an existing six-wire patch panel using a three-pair wiring system.

	AM-359 BOARD PAIR #	RJ-45		FEMALE RJ21 CONNECTOR
PORT 1	P1+	5	WHT/BLU	26
	P1-	4	BLU/WHT	1
	P2+	3	WHT/ORG	27
	P2-	6	ORG/WHT	2
	P3+	2	WHT/GRN	28
	P3-	7	GRN/WHT	3
PORT 2	P1+	5	WHT/BLU	29
	P1-	4	BLU/WHT	4
	P2+	3	WHT/ORG	30
	P2-	6	ORG/WHT	5
	P3+	2	WHT/GRN	31
	P3-	7	GRN/WHT	6
PORT 3	P1+	5	WHT/BLU	32
	P1-	4	BLU/WHT	7
	P2+	3	WHT/ORG	33
	P2-	6	ORG/WHT	8
	P3+	2	WHT/GRN	34
	P3-	7	GRN/WHT	9
PORT 4	P1+	5	WHT/BLU	35
	P1-	4	BLU/WHT	10
	P2+	3	WHT/ORG	36
	P2-	6	ORG/WHT	11
	P3+	2	WHT/GRN	37
	P3-	7	GRN/WHT	12

	AM-359 BOARD PAIR #	RJ-45		FEMALE RJ21 CONNECTOR
PORT 5	P1+	5	WHT/BLU	38
	P1-	4	BLU/WHT	13
	P2+	3	WHT/ORG	39
	P2-	6	ORG/WHT	14
	P3+	2	WHT/GRN	40
	P3-	7	GRN/WHT	15
PORT 6	P1+	5	WHT/BLU	41
	P1-	4	BLU/WHT	16
	P2+	3	WHT/ORG	42
	P2-	6	ORG/WHT	17
	P3+	2	WHT/GRN	43
	P3-	7	GRN/WHT	18
PORT 7	P1+	5	WHT/BLU	44
	P1-	4	BLU/WHT	19
	P2+	3	WHT/ORG	45
	P2-	6	ORG/WHT	20
	P3+	2	WHT/GRN	46
	P3-	7	GRN/WHT	21
PORT 8	P1+	5	WHT/BLU	47
	P1-	4	BLU/WHT	22
	P2+	3	WHT/ORG	48
	P2-	6	ORG/WHT	23
	P3+	2	WHT/GRN	49
	P3-	7	GRN/WHT	24

PDB-00359-51 External Cabling Kit

PDB-00359-52

Use this kit to replace an AM-358 I/O board with an AM-359 board when connecting to an existing eight-wire patch panel using a four-pair wiring system. This kit creates two extra eight-wire AM-355 type ports (ports 7 and 8), which have standard female DB9 connectors.

	AM-359 BOARD PAIR #	RJ-45		FEMALE RJ21 CONNECTOR
PORT 1	P1+	5	WHT/BLU	26
	P1-	4	BLU/WHT	1
	P2+	3	WHT/ORG	27
	P2-	6	ORG/WHT	2
	P3+	2	WHT/GRN	28
	P3-	7	GRN/WHT	3
	P4+	8	WHT/BRN	29
	P4-	1	BRN/WHT	4
PORT 2	P1+	5	WHT/BLU	30
	P1-	4	BLU/WHT	5
	P2+	3	WHT/ORG	31
	P2-	6	ORG/WHT	6
	P3+	2	WHT/GRN	32
	P3-	7	GRN/WHT	7
	P4+	8	WHT/BRN	33
	P4-	1	BRN/WHT	8
PORT 3	P1+	5	WHT/BLU	34
	P1-	4	BLU/WHT	9
	P2+	3	WHT/ORG	35
	P2-	6	ORG/WHT	10
	P3+	2	WHT/GRN	36
	P3-	7	GRN/WHT	11
	P4+	8	WHT/BRN	37
	P4-	1	BRN/WHT	12
PORT 4	P1+	5	WHT/BLU	38
	P1-	4	BLU/WHT	13
	P2+	3	WHT/ORG	39
	P2-	6	ORG/WHT	14
	P3+	2	WHT/GRN	40
	P3-	7	GRN/WHT	15
	P4+	8	WHT/BRN	41
	P4-	1	BRN/WHT	16

	AM-359 BOARD PAIR #	RJ-45		FEMALE RJ21 CONNECTOR	
PORT 5	P1+	5	WHT/BLU	42	
	P1-	4	BLU/WHT	17	
	P2+	3	WHT/ORG	43	
	P2-	6	ORG/WHT	18	
	P3+	2	WHT/GRN	44	
	P3-	7	GRN/WHT	19	
	P4+	8	WHT/BRN	45	
	P4-	1	BRN/WHT	20	
PORT 6	P1+	5	WHT/BLU	46	
	P1-	4	BLU/WHT	21	
	P2+	3	WHT/ORG	47	
	P2-	6	ORG/WHT	22	
	P3+	2	WHT/GRN	48	
	P3-	7	GRN/WHT	23	
	P4+	8	WHT/BRN	49	
	P4-	1	BRN/WHT	24	
PORT 7				FEMALE DB9	
	P1+	5	WHT/BLU	2	GRN
	P1-	4	BLU/WHT	5	RED
	P2+	3	WHT/ORG	3	BLK
	P2-	6	ORG/WHT	9	YEL
	P3+	2	WHT/GRN	4	ORG
	P3-	7	GRN/WHT	7	BRN
	P4+	8	WHT/BRN	8	SLT
	P4-	1	BRN/WHT	1	BLU
PORT 8				FEMALE DB9	
	P1+	5	WHT/BLU	2	GRN
	P1-	4	BLU/WHT	5	RED
	P2+	3	WHT/ORG	3	BLK
	P2-	6	ORG/WHT	9	YEL
	P3+	2	WHT/GRN	4	ORG
	P3-	7	GRN/WHT	7	BRN
	P4+	8	WHT/BRN	8	SLT
	P4-	1	BRN/WHT	1	BLU

PDB-00359-52 External Cabling Kit

PDB-00359-53

Use this kit to connect a modem to the AM-359 board with full handshaking support.

AM-359 BOARD			PARALLEL PATCH CORD	MALE DB25 ADAPTER = MODEM			
SIGNAL	PAIR #	RJ-45		RJ-45		DB25 PINOUTS	SIGNAL
RXD	P1+	5	WHT/BLU	5	GRN	3	TXD
RTS	P1-	4	BLU/WHT	4	RED	4	CTS
TXD	P2+	3	WHT/ORG	3	BLK	2	RXD
DTR	P2-	6	ORG/WHT	6	YEL	20	DTR
CTS	P3+	2	WHT/GRN	2	ORG	5	RTS
GND	P3-	7	GRN/WHT	7	BRN	7	GND
DCD	P4+	8	WHT/BRN	8	SLT/WHT	8	DCD
SHLD	P4-	1	BRN/WHT	1	BLU	1	SHLD

PDB-00359-53 External Cabling Kit

PDB-00359-54

Use this kit to connect all terminals and printers to the AM-359 board.

AM-359 BOARD			PARALLEL PATCH CORD	MALE DB25 ADAPTER = CRTs/PRINTERS			
SIGNAL	PAIR #	RJ-45		RJ-45		DB25 PINOUTS	SIGNAL
RXD	P1+	5	WHT/BLU	5	GRN	2	TXD
RTS	P1-	4	BLU/WHT	4	RED	5	CTS
TXD	P2+	3	WHT/ORG	3	BLK	3	RXD
DTR	P2-	6	ORG/WHT	6	YEL	8	DCD
CTS	P3+	2	WHT/GRN	2	ORG	20	DTR *
GND	P3-	7	GRN/WHT	7	BRN	7	GND
DCD	P4+	8	WHT/BRN	8	SLT/WHT	4	RTS
SHLD	P4-	1	BRN/WHT	1	BLU	1	SHLD

* CRT or printer busy

PDB-00359-54 External Cabling Kit

Both of the above cable kits require a shielded parallel patch cord with RJ-45 connectors at each end, like the 10 foot PRA-00189-10 cables supplied in the PDB-00359-56 cabling kit.

PDB-00359-55

Use this kit to connect a PC-AT style serial port to an AM-359

AM-359 BOARD			PARALLEL PATCH CORD	FEMALE DB9 ADAPTER = PC-AT			
SIGNAL	PAIR #	RJ-45		RJ-45		DB9 PINOUTS	SIGNAL
RXD	P1+	5	WHT/BLU	5	GRN	3	TXD
RTS	P1-	4	BLU/WHT	4	RED	8	CTS
TXD	P2+	3	WHT/ORG	3	BLK	2	RXD
DTR	P2-	6	ORG/WHT	6	YEL	1	DCD
CTS	P3+	2	WHT/GRN	2	ORG	7	RTS
GND	P3-	7	GRN/WHT	7	BRN	5	GND
DCD	P4+	8	WHT/BRN	8	SLT/WHT	4	DTR
SHLD	P4-	1	BRN/WHT	1	BLU	nc	no connect

PDB-00359-55 External Cabling Kit

The above cable kit requires a shielded parallel patch cord with RJ-45 connectors at each end, like the 10 foot PRA-00189-10 cables supplied in the PDB-00359-56 cabling kit.

Appendix B - Lightning and Overvoltage Protection

The issue of protection devices for communication lines associated with computer equipment is an area of growing concern. The increased sensitivity of electronic devices to electrical interference is the driving force behind this concern.

The protection of communication circuits in computer based equipment requires the use of a device to transfer data without signal degradation, and yet rapidly and safely limit power transients and surges. If the transients and surges are not limited in amplitude and rapidly removed from the data line circuit, the data drivers and receivers will be damaged or destroyed. If they are damaged, intermittent operation and loss of data integrity may result.

Even inside a building, an adjacent metal pipe or electrical conduit may carry electrical current which can induce current into the data line. The data line, which is seldom protected by its own conduit, is very susceptible to this condition. This rule even applies when running long cables (100 feet or longer) in one building either horizontally or vertically in elevator or cable shafts. In all such cases the communication cable should have a protector installed.

Even electrical faults within a facility will generate heavy surge currents. An external lightning surge or utility fault can exaggerate this condition. Under these circumstances, small residual ground loops can turn into catastrophic data line surges. While a direct lightning strike to a building may be regarded as a rare occurrence, the example is still quite valid. Even a fast moving cold front (many low clouds and high winds) can cause a great voltage potential within a facility.

Lightning strikes may be attracted by a chimney, external antenna, or an outside light. In any event, the lightning pulse will follow any path in its pursuit of “ground.” Any resistance in these paths will develop a voltage directly proportional to the current in the pulse. Data lines are no exception and their relatively high resistance leads to the development of significant voltages. (24 awg Copper wire = 2.5ohms/100 ft.)

A direct lightning strike does not have to hit a building to induce a potential. The building acts like an antenna. Every lightning pulse creates expanding fields of energy. The energy transmitted by a 100,000 ampere lightning strike can induce three volts per meter at a distance of 10 Kilometers and 100 volts per meter for a close strike. This type of energy field can even create a potential from one side of the building to the opposite side. The data lines run within the building provide perfect paths for this unwanted energy.

The role of the data line protector is quite simple. The device must keep voltages at sending and receiving stations below 24 volts despite several thousand volt induced potentials, and divert unwanted surge currents of several hundred to several thousand amperes away from the computer communication interface.

The protection device must clamp at the RS232 levels so the drivers and receivers are not damaged. This device must not break down over time or the clamping level increase due to age or heating. Many protection devices on the market today use metal oxide varistors (MOV) with slower, non-linear

response characteristics. This results in a protector which survives an impulse without protecting the data circuit. Additionally, the MOV is always forced to conduct through a resistance. This leads to heat build-up inside the device, and develops voltage potential across the device proportional to the current flowing through the internal resistance. This combination limits the ability of the MOV to clamp heavy current pulses without passing large voltage levels on to the protected load.

The protection device that Alpha Micro believes to be most effective is the silicon avalanche diode. This device is designed to avalanche and clamp at 18 volts for RS232. The silicon avalanche diode is precisely manufactured to achieve a bulk breakdown (avalanche) when the voltage exceeds this pre-determined level. Silicon avalanche diodes are intended to be current shunt devices, whereas MOVs are intended to be resistive shunt devices. The silicon avalanche diode is not intended to absorb the current, but rather divert the energy to ground, whereas the MOV is intended to absorb and dissipate energy by heat. The silicon avalanche RS232 protector effectively works in parallel with the signal line.

The silicon avalanche diodes possess numerous advantages over the MOV. This type of diode does not lose current handling ability with repeated pulses, as the MOV does. The silicon unit will clamp voltage and current surges to safe levels, while the MOV may not. The silicon diode clamps rapidly, typically 10 nanoseconds, while the MOV may not due to heating. The MOV increases in resistance, increasing its breakdown voltage over time, while the silicon diode does not. The MOV when it fails will open, leaving the data line circuit unprotected. The silicon avalanche diode rarely fails, but when it does it fails in a shorted condition. This will tell the user the device needs replacement because the data line circuit will not pass any signal, requiring the user to investigate.

CYLIX LIGHTNING PROTECTION DEVICE FOR USE WITH THE AM-359

Cylix Corporation has developed two types of protection devices that will work with the AM-359 board. The first model, DDP-H4, protects 32 wires (four AM-359 serial ports) and has four RJ-45 input connectors and four RJ-45 output connectors. The second model, DDP-H8, protects 64 wires (eight AM-359 serial ports) and has eight RJ-45 input connectors and eight RJ-45 output connectors. Both of the units are very easy to install. One end of the protection device has, depending on the model, either four or eight short cables with RJ-45 connectors. These cables plug directly into the AM-359 board. The protection device has a green ground wire that must be attached to the rear panel of your computer, which will tie the drain wire to chassis ground. After that, it is simply a matter of plugging the I/O cables you want to protect into the opposite end of the protector.

For ordering information please contact:

Cylix Corporation
2637 Townsgate Road, Suite 200
Westlake Village, Ca. 91361
1- 805 -379 - 3155

Appendix C - AM-905(-20) I/O Expansion Subsystem

The AM-905(-20) I/O subsystem is designed to support up to 10 AM-359 boards (80 serial ports). The product installation kit for the AM-905(-20) is shipped with all necessary internal and external cabling, including an adapter used for getting your 60-pin I/O cable out of the main computer.

When using the AM-905 subsystem, all the I/O boards connected to your 60-pin AM-355 I/O bus connector will need to be transferred from your computer's rear panel to the AM-905 subsystem. However, if your computer has other serial I/O boards connected to an AM-350 board, those boards can remain connected to your computer's rear panel. Also, if you have an AM-3000M or AM-4000 computer with dual 60-pin I/O connectors, one of these connectors can be used to support I/O boards connected to your computer's rear panel, while the other connector is used to support I/O boards installed in the AM-905 external subsystem.

AM-905(-20) I/O BOARD INSTALLATION

The process of installing I/O boards in the AM-905 subsystem is basically the same as the procedure for installing I/O boards in your host computer:

1. Each I/O board is secured to the AM-905's rear panel using three #6-32 Phillips-head screws.
2. The 60-pin cable that links each I/O board together is already installed in the AM-905 subsystem; all you need to do is plug the 60-pin cable into each board. Remember, all cable connections must be made pin-1 to pin-1. When installed correctly, the 60-pin cable will pass from board to board with no twists.

   Because the 60-pin cable carries power, it must not be twisted (reversed) as it is sequentially connected to each I/O board. If it is twisted, serious circuit damage will occur.

3. The AM-905 subsystem also has an auxiliary +5 volt power cable, similar to the one described earlier in this document. This cable is designed to prevent a power drop on the 60-pin I/O cable when six or more I/O boards are used. The auxiliary power cable has three +5 volt power connectors designed to plug directly into the AM-359 board. Plug one of the +5 volt connectors into the AM-359 board located farthest away from the AM-349 board. The remaining +5 volt connectors should be plugged into two other AM-359 boards located closer to the AM-349 board. The objective is to evenly distribute the +5 volt power between the AM-359 boards on the 60-pin I/0 cable. See the section “Preventing DC Power Drops on the I/O Bus” for more information on how to properly install the +5 volt auxiliary power cable.
4. Once all the I/O boards have been completely installed in the AM-905 subsystem, a shielded, 60-pin cable is used to connect the AM-905 subsystem to your host computer. This external cable is included with your AM-905(-20) subsystem.

5. Once the AM-905 subsystem is cabled to your host computer, you will need to follow the instructions in the section “Defining the Ports to the Operating System,” which appear earlier in this document.